



SEAT No. _____

[141]

No. of Pages: 02

SARDAR PATEL UNIVERSITY
BCA (IV - Semester) Examination
Wednesday, 06th April 2022
03.00 P.M. – 05.00 P.M.
Data Structure - II (US04CBCA25)

Total Marks: 70

Q-1 Multiple Choice Questions:

10

1. A binary tree has at most _____ child.
A. Two. B. Three.
C. One. D. None of these.
2. _____ is the first in the hierarchical arrangement of data items of a tree.
A. Branch. B. Parent.
C. Root. D. Leaf.
3. Linked list is _____ type of data structure.
A. Linear B. Non-Linear
C. Both (A) and (B) D. None of the Above
4. A type of the linked list in which none of the node contains NULL pointer is?
A. Singly linked list B. Doubly linked list
C. Circular linked list D. None of the Above
5. _____ is the process to arrange the data in ascending or descending order.
A. Sorting B. Inserting
C. Searching D. Updating
6. Merge sort uses...
A. Divide and Conqueror B. Dynamic programming
C. Greedy D. Backtracking
7. _____ file is also known as Random file.
A. Serial B. Direct
C. Sequential D. None of these
8. A _____ which defines a mapping from key space to the address space.
A. Deblocking B. Blocking
C. Hashing Function D. None of these
9. The lowest level of index is _____.
A. Track index B. Prime index
C. Index area D. Master index
10. _____ of tree is the maximum level of any node in the tree.
A. Depth B. Width
C. Height D. None of These

Q-2 Fill in the Blanks & True / False.

08

1. _____ is the maximum level of any node in the tree. (Depth of tree, Breadth of tree)
2. Leaf node is known as _____. (terminal node, binary node)
3. The minimum number of fields with each node of double linked list is _____. (3,2)
4. A _____ list in which the pointer points only to the successor node is singly linked list. (linear, double)
5. Algorithm that uses tape or disk during the sort is an external sorting algorithm. (True/False)
6. Binary searching technique, elements are eliminated by half in each pass. (True/False)
7. A Collision-resolution technique resolves conflicts that arise when more than one record key is mapped to the same table location. (True/False)
8. The Index area is created automatically by the data management routines in the operating system. (True/False)

①

P.T.O.

Q-3 Short Answer Questions (Any 10 out of 12).

20

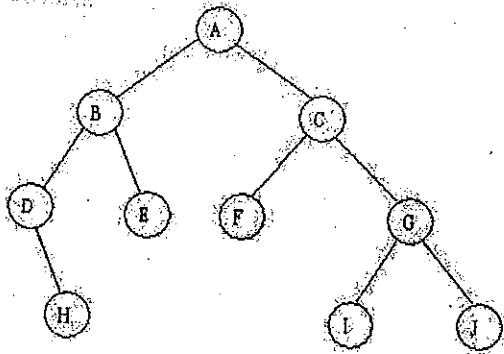
1. Define Binary Tree. List different types of binary tree.
2. Define Traversals. List the types of binary tree traversal.
3. Define Following Terms: Tree, Root.
4. Explain linked list in brief.
5. Lists insert operations that can be performed on singly linked list.
6. Explain circular linked list in brief.
7. List out Categories of Sorting. Define any one of them.
8. Advantages and Disadvantages of Linear search & Binary Search.
9. Trace the given data using Bubble sort: 5, 1, 4, 2, 8.
10. Define: Hashing function, Collision resolution technique.
11. List the types of overflow are used in indexed sequential file organization.
12. Define Bucket Capacity.

Q-4 Long Answer Questions (Attempt Any 04 out of 08).

32

1. Lexically ordered binary tree. Write a detail note on insertion and deletion operations performed on lexically ordered binary tree.
2. Traverse the given tree using Inorder, Preorder and Post order traversals

Given tree:



3. Write an algorithm to insert node at front position in singly linked list.
4. Write an algorithm to delete node from any position in singly linked list.
5. What is sorting? List sorting techniques and explain ANY ONE of them with algorithm.
6. What is searching? List searching techniques and explain ANY ONE of them with algorithm.
7. Write a detail note on structure and processing of index sequential file supported by IBM.
8. List and explain popular hashing function with example.

— X —